Original Article

Choroidal melanoma: A review of the experience of the Sydney Eye Hospital Professorial Unit 1979–1995

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Abstract

Purpose: This study examines 90 patients presenting with choroidal or ciliochoroidal melanoma to the Professorial Unit at the Sydney Eye Hospital. The indications for treatment, and the outcome for the eye and vision are presented together with an account of mortality and the incidence of metastases.

Methods: A retrospective analysis of 90 choroidal melanoma patients managed by one surgeon over a 16-year period was undertaken. Initial findings, investigations performed, incidence of metastatic disease, treatment received and complication rates and mortality, where applicable, were recorded.

Results: The group was followed for an average of 64 months (range, 5–172 months). Primary treatment was with either Iodine125 (^{125}I) brachytherapy, local excision or enucleation. Radiation retinopathy was prominent in ^{125}I cases resulting in poor visual acuity when the tumour resided in the posterior pole. Local excision even of large tumours was effective particularly if peripheral. Overall metastatic disease was seen in 11% with 5-year survival rates for the metastatic group being 10%. Prognosis after diagnosis of metastases was poor.

Conclusions: Specific therapy for choroidal melanoma must relate to the size and location of the tumour at the time of diagnosis. Visual outcome relates directly to the proximity of the tumour to the optic nerve and fovea. Metastatic disease latency can be prolonged; therefore caution about prognosis is required long after therapy is given. The 5-year survival is encouraging with all forms of therapy. However, as the natural history of ocular melanoma is variable and not fully delineated it is important to monitor the effects of conservative therapy. Further long-term survival data is required to distinguish whether one form of treatment is advantageous over the others, although case-control studies are difficult for ethical and practical reasons. In this regard the Collaborative Ocular Melanoma Study (COMS) will provide further evidence for the safety and efficacy of conservative therapy with brachytherapy compared to enucleation.

Key words: brachytherapy, enucleation, local resection, melanoma mortality, uveal melanoma.

Malignant melanoma of the choroid is the most common primary ocular malignancy in adults, occurring in approximately 6–7 cases per million people per year according to US figures.[1] This translates to 90–100 cases in Australia each year; the actual incidence rate in Australia is currently under review.

It is the main primary intraocular disease that can be fatal in adults; indeed, studies have shown that tumour-related mortality can range from 16% for small tumours up to 53% for large tumours over a 5-year period.[2] Other studies over longer periods of time have varied from 25% mortality over 15 years[3] to more than 50% in one study carried out over 25 years.[4] These figures refer particularly to mortality rates following enucleation.

The incidence of metastatic disease, usually to the liver (up to 95% in some cases[5]), also generates great interest as current treatment regimens for metastatic choroidal melanoma are not particularly successful. One study found that the median survival was 2 months on no treatment and 5.2 months for those on therapy.[5] Therefore metastatic disease in this context is a terminal disease and once it is clinically apparent prognosis is poor. It must also be recognized that although metastatic disease generally appears within 5 years of diagnosis cases may present up to 30 years later.[6]

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Accordingly, a great deal of attention is given to this potentially fatal condition in the literature particularly relating to the treatment options and the morbidity and mortality outcomes. Current management of malignant melanoma of the uvea remains controversial with the aim of treatment being local excision, destruction of the tumour and preservation of vision without inducing metastases or local side effects; this is difficult to achieve.[7]

Traditionally enucleation was the principal treatment as soon as diagnosis had been established. This was challenged by Zimmermann and McLean who argued that enucleation leads to iatrogenic tumour dissemination with peak mortality 2 years after surgery.[8] Their study has been criticized because of loss of patients to follow-up and mortality figures were not purged of other causes of death. Nevertheless Zimmermann's paper led to an increase in conservatism and to the introduction of other treatment modalities depending on the size and extent of the tumour plus the tumour location. These include radiotherapy (both brachytherapy and teletherapy), local excision, laser photocoagulation and cryotherapy.

Radiotherapy can be given via episcleral plaque or charged particle irradiation and the relative indications are similar.[9] Charged particle therapy offers theoretical advantages of better localization and greater uniformity of dose; however, plaque therapy is favoured as it is associated with less severe local complications, particularly those of the anterior segment.[9,10] Plaque therapy can be given for tumours up to 15 mm in diameter and a notched plaque has been developed for those adjacent to the optic nerve. More recently the Iodine^{125} (^{125}I) and Ruthenium plaques have taken over from the Cobalt^{60} plaques in most institutions. Radiotherapy using Cobalt^{60} plaques produced initial encouraging results in the short term and were associated with a 5-year survival similar to those treated with enucleation.[10–13] The cases in these studies were not randomized and it is unclear how many cases were regarded as unsuitable for radiotherapy treatment and not included in the study.

Local excision by partial lamellar sclerouvectomy[14–16] is a useful alternative particularly for peripheral tumours including those involving the ciliary body. A special form of local resection is endoresection, which can be used for residual or recurrent tumour.[17] On current evidence there is little difference in patient survival between local resection and enucleation or radiotherapy.[10,18–21]

Photocoagulation has been used for treating smaller tumours, those less than 3 mm thick and more than 3 mm from the fovea. It is used less frequently today as it requires multiple treatment sessions and can lead to vitreoretinal traction with visual distortion.[9] In general this form of treatment has been replaced by radioactive plaque therapy.

Withholding treatment is indicated if there is no evidence of activity or progression and particularly if the tumour is small; in such cases observation alone with regular review of tumour size and activity by fundus photography and ultrasound is regarded as an adequate measure.[22] Suggestive features of activity such as the presence of orange pigment (lipofuscin) on the tumour surface, elevation of more than 2.5 mm, subretinal fluid and proximity to the optic nerve need to be considered in the context of tumour size and progression.[10]

Overall, enucleation is still necessary for many large melanomas and those involving the optic nerve and beyond, while episcleral plaque brachytherapy has become the principal method for treating medium-sized and some large melanomas. These two methods of treatment are being currently assessed, particularly with regard to the long-term patient survival rates, in a multi-centre, randomized, prospective controlled trial, involving approximately 40 centres across the USA, known as the Collaborative Ocular Melanoma study (COM).[23]

The study reported here is the largest series to date in Australia reporting choroidal melanoma under the supervision of one surgeon. The study reviews the incidence of metastatic disease, mortality, and visual outcome with each of the various treatment modalities used.

Methods

A retrospective study set out to review the experience of the University Department of Ophthalmology, University of Sydney, under one surgeon (F.A.B.) over a 16-year period. We reviewed the records of all patients with a documented diagnosis of choroidal melanoma seen between August 1979 and May 1995. These records were obtained via computer search of the hospital patient records and the Professorial Unit records for patients coded for choroidal melanoma. This was cross-referenced with the NSW cancer registry records by searching for all cases of choroidal melanoma diagnosed and treated at the Sydney Eye Hospital.

Initial examination included presenting features (visual loss, field loss, photopsia/floaters, asymptomatic where tumour diagnosis was made at routine examination), tumour size and location, tumour ultrasound, visual acuity and investigation for systemic metastatic disease. These investigations included liver function tests, chest X-ray and abdominal and cerebral computed tomography scans.

Follow-up data was obtained by review of regular assessments, based on fundal examination and B scan ultrasound, documented by the same ophthalmologist in each case. In selected cases contact was made with individual patients, relatives or the local medical officer to ascertain up-to-date information on the development of systemic metastases and determining whether mortality was due to the tumour.

Data were collected on various tumour characteristics, patient presentation, treatment, response, complications and ultimate outcome both in visual terms and mortality. Tumour diameter and height as observed by indirect ophthalmoscopy were recorded, with confirmation where possible by B scan ultrasound.

The size and location of the tumour mass is important in deciding on the form of therapy and the ultimate outcome of the case. Tumour size was categorized into small, medium or large, following the guidelines in the COM study.[23,24] Tumours were: (i) small if the height was less than or equal to 3 mm and the base diameter was less than or equal to 8 mm; (ii) medium if the height was 3.1–8.0 mm and/or base diameter 8.1–16 mm; and (iii) large if the height was greater than or equal to 8.1 mm and/or the base diameter was greater than or equal to 16.1 mm.

Location of the tumour mass was recorded with reference to the optic disc and fovea. They were then categorized: (i) adjacent to the optic disc (within 3 mm); (ii) at the posterior pole (within 6 mm of the fovea); (iii) at the equator; and (iv) those peripheral (anterior to the equator).

Treatment modalities and the time to diagnosis were recorded in addition to complication rates and need for re-treatment. Overall three main therapeutic groups were identified: enucleation, ^{125}I brachytherapy, and local resection. In cases of enucleation this was also compared with the pathology examination findings. Histopathological assessment based on the modified Callender classification identified the cell type for the enucleation cases.[25]

The following treatment guidelines are used in general by the Professorial Unit.

(1) Observation for small tumours without suspicious features as outlined earlier.

(2) Enucleation for large tumours particularly if adjacent to the optic disc.

(3) Plaque radiotherapy for medium sized tumours up to 15 mm in basal diameter.

(4) Local resection for medium-sized tumours seen in the periphery.

A favourable treatment outcome was defined as one with no sign of metastases and visual outcome within 2 lines on the Snellen chart (except enucleation). For radiotherapy cases this included signs of tumour regression as determined by observation of tumour size, B scan ultrasound determining height and fluorescein angiographic evidence of atrophy and loss of tumour circulation. Parameters of active tumour regression included loss of lipofuscin and reduction in subretinal fluid.

Results

There were a total of 90 patients in the study with follow-up data to May 1995 or date of death available in 80 cases, with the rest studied for varying periods before being lost to follow-up. The follow-up period ranged from 5 to 172 months with a median follow-up of 64.1 months for the cases available for follow-up. For the 10 patients lost to follow-up the average follow-up time was 31.3 months; these patients have been excluded from follow-up analysis but included in analysis of treatment selection and early side-effects of therapy. All cases were unilateral with no systemic metastatic disease apparent at the time of diagnosis. This is in keeping with findings of other studies in which the incidence of metastases at diagnosis is also reported as low, approximately 0.5–3.5%.[10] The age at diagnosis ranged from 26 to 83 years, with a median of 57.5 years. This and the presenting features, as shown in Table 1, are similar to previous studies.[26] These features in themselves are insufficient to make a definitive diagnosis of choroidal melanoma. A large proportion of the cases, (60 of 90 (66%)), were located adjacent to the optic disc or at the posterior pole, which had implications for visual outcome after therapy. This is consistent with other studies that found the majority of cases involved the posterior pole.[27]

Treatment

The treatment received by the group was as follows: 7 (8%) observation; 20 (22%) enucleation; 43 (48%) with ^{125}I plaque

Table 1. Clinical parameters pertaining to choroidal melanoma patients

Variable	No.	%
Age (years)		
<30	2	2
30–40	10	11
40–50	16	18
50–60	24	27
60–70	21	23
>70	17	19
Sex		
Male	41	46
Female	49	54
Laterality		
Right	46	51
Left	44	49
Symptoms		
Asymptomatic	29	32
Visual acuity	36	40
Visual field loss	6	7
Photopsia	19	21
Size		
Small	14	16
Medium	62	70
Large	14	16
Location		
Adjacent to disc	35	39
Macula	27	30
Equator	15	17
Periphery	13	14

Table 2. Size and location of tumour mass in relation to therapeutic modality employed

	Size parameters			Location			
Therapy	Average volume index	Category	Total number	Adjacent to optic nerve*	Posterior pole	Equator	Periphery
Local resection	72.00						
		Small	—	—	—	—	—
		Medium	12	3	2	—	7
		Large	4	1	2	—	1
^{125}I plaque	40.98						
		Small	8	5	1	2	—
		Medium	32	14	12	6	—
		Large	3	1	—	—	2
Enucleation	73.50						
		Small	—	—	—	—	—
		Medium	14	4	8	2	—
		Large	6	4	—	1	1

*Within 3 mm of nerve head.

radiotherapy; 16 (18%) with local excision; and the remaining 4 cases with cryopexy, laser photocoagulation, helium radiotherapy and Ruthenium plaque therapy. Prior to institution of therapy, 31 cases (34%) were observed for a period of time (overall observation time ranged from 5 to 172 months with an average of 24 months). Of these, 7 cases (8%) are still under observation with no requirement for therapeutic intervention without adverse effects (follow-up range, 9–172 months; average, 79.6 months). Table 2 shows the size and location parameters and the management strategy undertaken.

Additional management was required due to progression or local recurrence of tumour in 19 cases (21%) with two of these undergoing three individual sets of therapy. These treatment modalities and outcomes are indicated in Table 3.

Pathology

Where enucleation or local resection was performed (42 cases) all specimens were examined by the same pathologist at Sydney Eye Hospital. The breakdown of cell types was as follows: spindle A/B 12% (5 cases); spindle B 50% (21); mixed spindle and epithelioid 33% (14); and epithelioid 2% (1). In the remaining one case the cell type was not available.

Visual outcome

We compared visual acuity at presentation with visual acuity at final review. These values are summarized in Table 4 for the actual visual acuity and Table 5 for the comparison. Where the visual acuities were compared, a 'worse outcome' is defined as loss of 2 lines or more on the Snellen chart.

Complications of treatment modalities

Complications were primarily seen in cases treated with local excision and ^{125}I radiotherapy; enucleated cases were trouble-free apart from poor fit of prosthesis and some mucous discharge and socket problems in 6 cases (19%). Articulated scleral ball implants were used following all enucleations.

Table 3. Outcome for patients failing primary therapy

Initial therapy	Re-treatment	3rd treatment	*n*	Visual outcome	Mets.	Mortality
^{125}I plaque	Re-plaque with ^{125}I		3	2 worse*	2	1
	Enucleation		3	—	—	—
Local excision	^{125}I plaque		4	All same	—	—
	Enucleation		4	—	2	2
Laser photocoagulation	Enucleation		1	—	—	—
Helium radiotherapy	Enucleation		1	—	—	—
Cryopexy	Enucleation		1	—	—	—
Local excision	^{125}I plaque	Enucleation	1	—	—	—
^{125}I plaque	Re-plaque with ^{125}I	Enucleation	1	—	—	—

*A worse visual outcome is defined as loss of 2 lines or more on the Snellen chart.

Table 4. Visual outcome based on treatment modalities and tumour location

		6/6–6/18		6/24–6/60		5/60–CF		HM–LP		Enucl.
		IVA	FVA	IVA	FVA	IVA	FVA	IVA	FVA	FVA
^{125}I		29	11	11	6	3	13	—	9	4
LR		12	6	3	2	1	3	—	—	5
Enucleation		13	—	5	—	—	—	2	—	20
Helium RXT		1	—	—	—	—	—	—	—	1
Ruthenium RXT		1	1	—	—	—	—	—	—	—
Cryopexy		1	—	—	—	—	—	—	—	1
Laser		—	—	—	—	1	—	—	—	1
Observation		6	6	1	1	—	—	—	—	—
Total no. cases		63	24	20	9	5	16	2	9	32
Location										
ON	^{125}I	12	3	7	4	1	6	—	4	3
	LR	2	1	1	1	1	1	—	—	1
	E	3	—	3	—	—	—	2	—	8
	RRXT	1	1	—	—	—	—	—	—	—
	Laser	—	—	—	—	1	—	—	—	1
	OB	1	1	—	—	—	—	—	—	—
	Total	19	6	11	5	3	7	2	4	13
PP/M	^{125}I	8	1	3	1	2	6	—	4	1
	LR	3	2	1	—	—	—	—	—	2
	E	7	—	1	—	—	—	—	—	8
	HRXT	1	—	—	—	—	—	—	—	1
	CRYO	1	—	—	—	—	—	—	—	1
	OB	2	2	—	—	—	—	—	—	—
	Total	22	5	5	1	2	6	—	4	13
EQ	^{125}I	8	6	—	1	—	1	—	—	—
	LR	—	—	—	—	—	—	—	—	—
	E	3	—	—	—	—	—	—	—	3
	OB	2	2	1	1	—	—	—	—	—
	Total	13	8	1	2	—	1	—	—	3
P	^{125}I	1	1	1	—		—		1	—
	LR	7	3	1	1	—	2	—	—	2
	E	—	—	1	—	—	—	—	—	1
	OB	1	1	—	—	—	—	—	—	—
	Total	9	5	3	1	—	2	—	1	3

^{125}I, Iodine 125 brachytherapy; LR, local excision; ON, adjacent to the optic nerve (within 3 mm); PP/M, macular (within 6 mm); EQ, equatorial; P, peripheral; E, enucleation; HRXT, Helium radiotherapy; RRXT, Ruthenium radiotherapy; CRYO, cryopexy; OB, observation.

Iodine^{125} brachytherapy (43 cases) resulted in 28 cases (65%) developing radiation retinopathy or optic neuropathy. Retinopathy was common where the posterior pole or proximity to the optic nerve was involved. The tumour was located in the aforementioned area in 23 of the 28 cases. The time to development of retinopathy varied, ranging from 2 to 63 months with a median value of 18 months. Therefore, the risk of visual deterioration is ongoing after initial treatment particularly if there is retinopathy at the posterior pole. Cataract development, usually of the posterior subcapsular type, occurred in 7 cases (16%) and rubeotic glaucoma was seen in one case. This resulted from central retinal artery occlusion secondary to radiation retinopathy.

Local excision (16 cases) resulted in few complications directly attributable to surgery, with anterior segment ischaemia secondary to long ciliary vessel damage occurring in one case and vitreous haemorrhage in three other cases.

Local intraocular recurrence of disease was seen in 17 cases (21%) followed until study endpoint or death (total n = 80). The treatment modalities for cases with intraocular recurrence were: 5 cases of ^{125}I, 9 cases of local resection and one case each of laser photocoagulation, cryotherapy, and helium radiotherapy. The posterior pole was involved in 7 cases, reflecting the technical difficulties involved. Retreatment showed regression of tumour growth in all cases; despite this, four patients went on to develop systemic metastases. Evidence of arrested disease or regression following ^{125}I brachytherapy, as outlined in the Methods section, was seen in 88% of cases and considered a favourable outcome.

Metastatic disease

Metastatic disease was identified in 10 (13%) of the follow-up group. Only one patient remained alive at the study endpoint in May 1995. Table 6 indicates the features relating to these cases. The majority of cases metastasized to the liver with one interesting exception that involved bone. Also notable is the relatively long latent period to development of metastases but once diagnosis is made there is poor prognosis.

Mortality

The mortality rate from the study population was 16% (13 of 80) with the mortality rate directly related to choroidal melanoma being 11% (9 cases) due to metastatic choroidal melanoma. This is not statistically significant. The cause of death for the remainder was: two cases of myocardial infarction, one case of pancreatic carcinoma (unrelated) and one case who died overseas of unknown cause. The 5-year survival rates for the three main treatment modalities are as follows: enucleation 81%; ^{125}I radiotherapy 93%; and local resection 79%. However, the average size (measured as volume index = diameter × height)[3] of enucleated and locally resected tumours were larger than the plaqued cases: enucleation 73.50; local resection 72.00; and ^{125}I plaque 40.98.

A final interesting feature was the presence of other primary malignancies in seven cases; this has been documented elsewhere.[28] There was no particular trend and they are listed as possibly other mutations in the *p53* oncogene. Primary melanoma elsewhere (foot, thigh, and abdominal wall) was seen in three cases, and one case each of bowel carcinoma, cerebral cancer, breast and parotid malignancy.

Discussion

The present series of choroidal melanoma was studied over a considerable period and in that time surgical procedures and therapeutic modalities have changed. However, this series provides a good indication of prognosis and visual outcome for this condition.

Taking the group as a whole the overall therapeutic success rate in terms of survival was good (89%). Where multiple therapy was required 68% of cases resulted in a good outcome both visually and systemically. Intraocular disease recurrence as opposed to relapse occurred in 19% of the total study population, which compares to other studies.[17] The incidence of metastatic disease (local, orbital and systemic) and survival rate at 5 years is comparable to other studies.[9,10,13,29] Indeed, survival rates for the treatment groups were similar; demonstrating a good systemic outcome for each treatment method. Complications of the procedures in each group were ^{125}I 65%, local resection 25% and enucleation 30%.

Table 5. Comparison of pretreatment and final assessment visual acuity outcome as recorded on Snellen chart

	Total		^{125}I		LR	
Location	S	W	S	W	S	W
ON	9	13	3	13	3	—
PP/M	5	10	2	11	1	—
EQ	7	4	3	4	—	—
P	6	4	1	1	4	3

ON, adjacent to the optic nerve (within 3 mm); PP/M, macular (within 6 mm); EQ, equatorial; P, peripheral; S, same; W, worse.

Observation

On the basis of the previously listed indications 31 cases were initially observed with 24 progressing to therapy. The remaining 7 continuing under surveillance at the time of this report had small tumours, as per the COM study definition[23] and have been observed for an average period of 79.4

Table 6. Outcome for cases with metastatic disease

Site	Time to metastases (months)	Life span	Diameter (mm)	Height (mm)	Therapy	Cell type
Liver	84	Alive	8	3	^{125}I/^{125}I	NA
Liver	12	3	11	4	^{125}I/^{125}I	NA
Liver	41	22	11	4	Enucl.	SB
Liver	37	5	11	5	LR	SB
Liver/lung	94	3	15	9	LR/Enucl.	SB
Liver	71	4	10	5	^{125}I	NA
Liver	92	3	8.5	3.5	^{125}I	NA
Liver	6	2	13	13	Enucl.	M
Liver	65	8	15	9	Enucl.	SB
Bone	70	6	10	5	LR/Enucl.	M
Av.	57.2	6.2				

^{125}I, Iodine 125 brachytherapy; LR, local excision; Enucl., enucleation; M, mixed spindle cell and epithelioid cell; SB, spindle B; Life span, life span after diagnosis of metastases.

months. At the completion of the review none of the group had shown progression of disease, deterioration in visual function or signs of metastatic disease. It appears that observation is appropriate for this specific group of small tumours and our findings are supported by Gass[22] and Augsburger.[30] However, we must consider that enlargement of the tumour may be potentially associated with a poorer outcome and that we may be seeing the same patient at a later phase in his inevitable course to the development of metastatic disease if it is going to occur.

Enucleation

With enucleation the outcome can be measured only by the survival rate or incidence of metastatic disease. In our group no case of local recurrence occurred and eye socket problems following the procedure were few, indicating success of the primary procedure. Overall the survival of 81% over 5.3 years is encouraging and comparable to other 5-year study outcomes whose average survival rates were 70–80%.[2,20,31,32] Longer term studies show deteriorating survival over time with average survival rates at 15 years of 43%.[4,33]

Metastatic disease occurred in three cases (15%) and these were associated with larger tumours (>11 mm diameter) with an average interval time to diagnosis of 37 months.

Local resection

This group is biased toward larger tumours where enucleation was recommended but refused by the patient and the tumour size and site were thought to be unsuitable for radiotherapy. The average volume index of tumours in this group was approximately 72.00 compared with that of the ^{125}I plaque tumours at 40.98 and the enucleated cases at 73.50.

The results of surgical resection for choroidal melanoma are related to the quality of vision retained and the incidence of complications (vitreous haemorrhage, retinal detachment and disease recurrence).[34,35] This depends to a large extent on the size and location of the tumour pre-operatively. Damato states that one of the most significant risk factors for visual loss is 'posterior tumour extension to within 1 disc diameter of the optic disc or fovea'.[36] This is mainly due to the technical difficulty with resection in these cases. Also vision retention appears to be better with nasally placed tumours.[34,36] One case of interest was a patient with a 16 mm tumour with extension into the vortex vein and who refused enucleation. Local resection with removal of the vortex vein *en-bloc* was required with scleral graft for repair. Visual results were good with 6/12 retained for 8 years before clinical evidence of metastases was seen. Overall there were fewer long-term complications if the initial operation was successful.[10]

Visual outcome compared favourably to other studies[18,19] with 8 (50%) of our cases showing a final visual acuity of 6/60 or better. This reflected the fact that most of the tumours with good visual outcome were located peripherally. Similarly, peripheral tumours treated with ^{125}I brachytherapy also did well visually as shown in Table 4 (80% of peripheral or equatorial tumours with a final visual acuity of 6/60 or better). Overall, 56% of our cases had a final visual acuity within 2 lines of the initial visual acuity.

Residual tumour and recurrence occurred in 56% and was weighted by tumours close to the optic disc. Experience now shows that local resection of these tumours is better not attempted. One case in particular appeared clear with visual acuity of 6/6 until 5 years later when retreatment was required. Laser was unsuccessful but a notched ^{125}I plaque caused complete resolution though it was associated with radiation optic neuropathy and a final visual acuity of 6/60. In spite of the need for adjuvant therapy in over 50% of the cases, prognosis and tumour control remains good in 78%. Retention of the globe is an important endpoint for this procedure in both visual and cosmetic terms and was achieved in 75% of cases. Overall survival post local resection was comparable to that of enucleation or ^{125}I plaque therapy with a 5-year survival of 79%. This is similar to other studies of local resection that showed 5-year survival rates of 79–84%.[18,19] Metastases occurred in 2 cases of 16 (13%).

Iodine125 brachytherapy

Local plaque radiotherapy is now an accepted and widely employed therapeutic modality for choroidal melanoma treatment, especially for medium-sized tumours.[27,37] The current problem is the unknown long-term survival rates compared to other forms of therapy, in particular enucleation.[38] The 5-year survival rate among this group in the present study was excellent with 93% of cases alive at the completion of the study, which compares favourably with enucleation studies[9,13] and with other ^{125}I studies (5-year survival rates of 83%).[29] Recurrence of disease was low, 5 cases of 43 (11%), and re-treatment resulted in a good systemic outcome in the majority. Regression of the primary tumour following ^{125}I therapy was also encouraging with 88% remaining static or showing signs of regression.

However, despite this outcome the incidence of complications, as with other studies of ^{125}I therapy,[16,29,39,40] was relatively high and this impacts mainly on the visual outcome rather than the survival outcome. In fact our data indicated significant loss of vision with only 28% of patients retaining a final visual acuity within 2 lines of the initial visual acuity. This is primarily due to the development of radiation retinopathy in the posterior segment, and is consistent with a study by Seddon et al. who showed that the probability of visual loss from radiation optic neuropathy or maculopathy doubles for tumours near to the optic disc or fovea compared to anterior tumours.[41] This obviously will be influenced by the tumour size and subsequent radiation dose received (tumour apex dose) by the normal ocular structures.[29] Patients feel vision preservation and cosmesis are the most important factors when considering therapy. With the loss of vision patients then come to realize that it is metastatic disease which is more of a threat. Therefore when vision deteriorates due to radiation-induced complications the justification for keeping the eye is weakened and enucleation may be the result (4 cases in our study (9%)). In a combined study of ^{125}I radiation plaques for all choroidal tumours treated in NSW the incidence of radiation retinopathy at the end of the first year was approximately 25%.[39] However, the lag time to development of retinopathy can be up to 5 years; therefore ongoing review post-therapy is important.

Larger posterior tumours are more at risk of retinopathy and subsequent visual prognosis is worse. This must be considered when selecting the method of therapy. Iodine^{125} classically involves a 100 Gy dose to the apex; however, studies have shown good tumour control with dosages as low as 70 Gy.[42] Studies of charged particle therapy with sequential reduction of dosage down to 50 Gy have also shown no differences in tumour control or complications.[43–45] The main problem remains that the tumoricidal radiation dose for choroidal melanoma is still not fully known.[29]

Packer et al. instituted the use of Palladium^{103} (27 keV), as a replacement for ^{125}I (35 keV). The inverse square law shows that with the increasing distance from the radiation source the energy transferred to tissues is substantially reduced. Therefore, a lower energy radionuclide source such as Palladium will further reduce the radiation dose to normal ocular structures via a greater fall-off rate. Long-term follow-up of this method of treatment is awaited. Ruthenium holds the same promise, particularly for management of residual tumours after local resection.

One unusual complication seen as a result of ^{125}I brachytherapy was the development of central retinal arterial occlusion (CRAO) approximately 9 months after therapy. This resulted in ischaemia and the development of rubeotic glaucoma with the eye subsequently becoming blind and painful requiring enucleation. This has been previously documented by Noble in a case of CRAO following external beam irradiation for Graves' thyroid ophthalmopathy.[46] He describes only two other cases of CRAO related to radiation therapy and our case is possibly the first to result following brachytherapy though the underlying pathogenesis is probably similar.

Metastatic disease was seen in 4 cases (9%) with an average interval time to diagnosis of 65 months.

Metastases

The natural history of unselected and untreated choroidal melanoma is unknown. When enucleation was routine the mortality figures for 5 years were approximately 25%.[2] Our study, which is inclusive of all patients presenting, has a 5 year mortality rate of approximately 10%.

Life expectancy is short once metastases are manifest; there is still no effective therapy for metastatic melanoma.[5,47–49] Our study confirms the liver as the major organ involved in metastases. Spindle B cell type was the most prevalent in the cases of enucleation and local resection. Consistent with other studies overall survival following the diagnosis of metastases was poor with an average life span of just 6.2 months despite therapy.[5]

Controversy about the natural history of the disease includes the notion that tumour presence may incite an immune response in the interest of patient survival. Further to this, areas of research are looking at this concept in relation to the development of metastases. Development in the animal model of tumour growth inhibition by the tumour mass via release of an angiogenesis inhibitor, angiostatin, has been studied.[50] This interesting concept is yet to be applied to human carcinogenesis theory, though there may be a subset of those individuals receiving multiple therapy for choroidal melanoma who may be adversely affected by a mechanism involving the tumour-mass inhibition effect on metastatic development via angiostatin. If this were true it would justify a more conservative approach but no conclusive basis can be found for it in the present study.

Conservative therapy of choroidal melanoma has resulted in more eyes retained and vision saved along with an increasing acceptance of therapy and patient satisfaction without a concomitant increase in mortality. The present study does show that posterior tumours have a poor prognosis for vision regardless of therapy. Those cases responding best to local resection were peripheral and/or those involving the ciliary body; therefore this would be the therapy of choice for this category of tumour. However, we must keep in mind that Greer et al.'s study showed that posterior tumours appear to have a better prognosis for life compared with anterior tumours.[3] The role of Ruthenium and Palladium in posterior tumours is deserving of further comparison with ^{125}I. It must be considered whether ^{125}I is as equally deleterious as the previously used Cobalt^{60} but over a longer time period.

More needs to be known about the natural history of choroidal melanoma and a greater understanding of factors concerned with pathogenesis of metastases such as angiostatin. This should include whether prognosis may be improved by therapies that may improve the patient's capacity to mount an immune response to the disease.

Conclusion

Choroidal melanoma is not a common disease but has the potential to be a rapidly fatal one after metastatic development. Of the 90 choroidal melanoma tumours followed for 16 years (mean, 5.2) in the study reported here there is no evidence that enucleation, local excision nor radiotherapy adversely affects outcome where they are compared to each other. The decision then rests on each individual case and must reflect the size and location of the tumour at the time of presentation. Systemic disease is not commonly seen at diagnosis but screening for liver disease and local extension must be looked for at the outset.

Enucleation is indicated when the site and size of the tumour precludes a reasonable expectation of complete surgical removal, effective irradiation or retained vision. The incidence of metastases after surgery in the present study did not support the theory of micrometastatic spread at the time of surgery or as a result of surgery later.

Local resection can be anticipated for tumours less than 16 mm in diameter that are located anterior to the major retinal arteriolar arcades including those involving the ciliary body. This method of treatment provides a good visual outcome free of radiation-induced complications and low recurrence rate when the tumour is not located posteriorly.

Iodine^{125} is now successful as a means of arresting progression of choroidal melanoma, as shown in the present study by a high regression rate. Good tumour response to therapy and excellent survival statistics make this a viable alternative but it is attended by high and unpredictable rates of sight-threatening radiation retinopathy particularly for tumours in the region of the posterior pole. Different isotopes such as Palladium or Ruthenium have been used to minimize radiation effects on normal ocular structures and further studies in this area to lower overall radiation dosage are necessary. It is important for patients with posteriorly placed tumours to have the risk of visual loss explained at the outset although the choice of plaque therapy is often made by the patients who usually prefer this to enucleation.

In terms of re-treatment the majority of cases respond well after either enucleation or use of the ^{125}I plaque. Where this is required favourable survival rates can be maintained for this group along with tumour regression. With few exceptions preservation of good central vision is poor when tumours come within 3 mm of the disc or macula. The possibility of combined therapy, in particular tumour bulk reduction by local resection followed by adjunctive plaque radiotherapy, should be considered. This may allow a lower dosage of radiation thereby reducing morbidity and avoiding enucleation for some larger tumours.

Overall we have found this disease responds well to therapy and survival after therapy is promising. The present study found complications and results were equivalent to other series and all clinicians should review their results to see if they are within the commonly accepted parameters of treatment. Metastatic disease is the most important issue relating to patient survival and large randomized series such as COMS will demonstrate the advantages of differing therapies with regard to patient survival. Given the latency of metastatic disease it is necessary to have long-term follow-up, which should include regular ophthalmic assessment in conjunction with systemic assessment by the local medical officer. In this way cases that have been stable for extended periods will not be lost to observation.

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